ANALYSIS REPORT

RCIM Model-Bank Reproduction SVM Micro-Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the dedicated `SVR` micro-closure campaign prepared in:

- `doc/reports/campaign_plans/track_1/svm/2026-04-14-21-42-47_track1_svm_micro_closure_campaign_plan_report.md`

  The campaign executed `8` exact-paper validation runs through the dedicated launcher:
- completed runs: `8`
- failed runs: `0`
- execution window: `2026-04-14 21:44:13+02:00` to `2026-04-14 21:52:37+02:00`
- campaign artifact root: `output/training_campaigns/track1/svm/track1_svm_micro_closure_campaign_2026_04_14_21_42_47/` The campaign goal was minimal and explicit:
- test only the last residual `SVM` harmonics `40`, `240`, and `162`;
- avoid reopening any already closed `SVM` surfaces;
- check whether one final narrow pass could finish the row.

Objective And Outcome

The campaign had three concrete goals

1. close amplitude `40`;
2. close amplitude `240` on `RMSE`;
3. close phase `162` on both `MAE` and `RMSE`.

   Outcome:

- all `8` runs completed successfully;
- all `8` runs exported ONNX artifacts with `0` failed exports;
- no campaign run materially improved the canonical `SVM` row beyond the

  current benchmark values;
- the campaign confirmed that the current `SVM` row is stable, but still not

  fully closed. This means the campaign succeeded operationally, but scientifically it acted as a confirmation pass rather than a closure pass.

Ranking Policy

This campaign is not a full-matrix family comparison. Each run only covers one scoped residual slice of the `SVM` row.

The serialized campaign policy is therefore

- primary metric: `cell_closure_score_desc`
- first tie breaker: `met_paper_cell_count_desc`
- second tie breaker: `open_paper_cell_count_asc`
- third tie breaker: `mean_normalized_gap_ratio_asc`
- fourth tie breaker: `max_normalized_gap_ratio_asc`
- fifth tie breaker: `improved_cell_count_desc`
- sixth tie breaker: `status_upgrade_score_desc`
- seventh tie breaker: `run_name`

  Where:
- `cell_closure_score = (met + 0.5 * near) / paper_cell_count`
- `met` means the scoped repository cell reached or beat the paper cell
- `near` means the positive gap stayed within `25%` of the paper cell
- `status_upgrade_score` rewards `red -> yellow`, `yellow -> green`, and `red -> green` transitions versus the pre-micro-closure `SVM` baseline row Interpretation:
- this policy exists only to serialize the required campaign-best artifacts;
- the real scientific interpretation remains the merged `SVM` row across Tables `2-5`.

Campaign Ranking

Ranked Completed Runs

Rank	Run	Scope	Cells	Met	Near	Open	ClosureScore	Improved	Upgrades
1	`track1_svm_amplitude_240_only`	`amplitudes_only`	2	1	1	0	0.750	0	0
2	`track1_svm_amplitude_micro_closure_seed23`	`amplitudes_only`	4	1	3	0	0.625	0	0
3	`track1_svm_phase_micro_closure_split15`	`phases_only`	2	0	2	0	0.500	0	0
4	`track1_svm_amplitude_40_only`	`amplitudes_only`	2	0	2	0	0.500	0	0
5	`track1_svm_amplitude_micro_closure_baseline`	`amplitudes_only`	4	0	4	0	0.500	1	0
6	`track1_svm_phase_micro_closure_baseline`	`phases_only`	2	0	2	0	0.500	0	0
7	`track1_svm_phase_micro_closure_seed23`	`phases_only`	2	0	2	0	0.500	0	0
8	`track1_svm_amplitude_micro_closure_split15`	`amplitudes_only`	4	0	3	1	0.375	1	0

Export Surface

Completed Runs	Exported ONNX Files	Failed Exports	Surrogate Exports
8	11	0	4

Campaign Best Run

The explicit bookkeeping winner is

- `track1_svm_amplitude_240_only`

It was selected because it achieved:

- the highest scoped cell-closure score: `0.750`
- `1` met cell and `1` near cell out of `2`
- the cleanest targeted result on the residual amplitude `240` slice

Important interpretation:

- this run is the campaign bookkeeping winner;
- it is **not** a new scientific row winner by itself;
- the canonical `SVM` row does not change as a result of this campaign.

SVM Row Impact

Table-Level Before Vs After

Surface	Before	After
Table `2` amplitude `MAE`	`9` green, `1` yellow, `0` red	`9` green, `1` yellow, `0` red
Table `3` amplitude `RMSE`	`8` green, `2` yellow, `0` red	`8` green, `2` yellow, `0` red
Table `4` phase `MAE`	`8` green, `1` yellow, `0` red	`8` green, `1` yellow, `0` red
Table `5` phase `RMSE`	`8` green, `1` yellow, `0` red	`8` green, `1` yellow, `0` red

Residual SVM Yellow Cells

- Table `2` : `40`
- Table `3` : `40` , `240`
- Table `4` : `162`
- Table `5` : `162`

There are still no remaining red `SVM` cells in Tables `2-5` .

Canonical Benchmark Impact

This campaign did not change the canonical benchmark totals.

The canonical benchmark therefore remains

- Table `2` full-matrix totals: `54 / 36 / 10`
- Table `3` full-matrix totals: `53 / 34 / 13`
- Table `4` full-matrix totals: `52 / 37 / 1`
- Table `5` full-matrix totals: `43 / 41 / 6`

  The harmonic-level Table `6` reading also remains unchanged.

Main Conclusions

The campaign supports four conclusions.

1. SVM Is At A Real Plateau

The micro-pass reproduced the current residual `SVM` quality but did not produce a further canonical upgrade.

2. The Remaining Gaps Are Very Small But Persistent

The residual cells `40`, `240`, and `162` stay in the yellow band, which means they are close but not yet fully closed.

3. Another Generic SVM Sweep Is Unlikely To Be Efficient

This campaign was already very narrow and still produced no canonical upgrade.

That is a practical sign that `SVM` is no longer the best place for another generic budget pass.

4. The Next SVM Work Should Be Either Surgical Or Stopped

The rational options are now

- one last highly surgical hand-designed `SVM` attempt on a single residual

  surface; or
- stop spending the main `RCIM Model-Bank Reproduction` budget on `SVM` and move to another row.

Produced Campaign Artifacts

The campaign output folder now contains the required serialized winner artifacts:

- `output/training_campaigns/track1/svm/track1_svm_micro_closure_campaign_2026_04_14_21_42_47/campaign_leaderboard.yaml`
- `output/training_campaigns/track1/svm/track1_svm_micro_closure_campaign_2026_04_14_21_42_47/campaign_best_run.yaml`
- `output/training_campaigns/track1/svm/track1_svm_micro_closure_campaign_2026_04_14_21_42_47/campaign_best_run.md`

  The canonical reports updated by this closure are:

- `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- `doc/reports/analysis/Training Results Master Summary.md`